

What Are the Smallest Building Blocks of Matter?

A Resolution-Ladder Program for Testing Pointlike Fermions

Anonymous Research Proposal

Abstract—The Standard Model describes quarks and leptons as elementary fermion fields and gauge bosons plus the Higgs boson as elementary bosonic fields, whereas protons and neutrons are composite hadrons. This classification is highly successful, but elementary has an experimentally limited meaning: scattering can test pointlike behavior over a finite range of momentum transfer and under stated model assumptions; it cannot prove absolute indivisibility at every scale. This evidence-bounded proposal develops Resolution-Ladder Model Comparison, a cross-channel program for distinguishing a pointlike Standard-Model explanation from finite-form-factor, effective contact-interaction, and excited-fermion alternatives. The framework versions the kinematic definition, selection, response model, and nuisance ledger for each channel; enforces an effective-field-theory validity gate; and evaluates models on held-out bins after adversarial tests of parton distributions, detector response, and theoretical truncation. The central hypothesis is that a shared resolution ledger yields more reliable discrimination of a coherent short-distance departure from artifacts of single-channel systematics than independent limits. The program is strictly a research design. It does not operate a collider, access restricted event records, generate a new exclusion, or claim a discovery.

Index Terms—Standard Model, elementary particles, quark compositeness, lepton compositeness, effective field theory, Drell-Yan, collider phenomenology, model comparison

I. INTRODUCTION

Everyday matter is assembled from atoms, but atoms are not the endpoint of physical description. Their electrons are described as leptons; their nuclei are built from protons and neutrons; and the latter are composite hadrons containing quark and gluon degrees of freedom. Particle physics goes further by describing quarks and leptons as matter fields and by describing gauge and Higgs fields as elementary bosonic sectors. This framework is the Standard Model (SM), a quantum field theory with gauge structure

$$SU(3)_C \times SU(2)_L \times U(1)_Y. \quad (1)$$

Particle-physics reviews compile the inventory, interaction conventions, and experimental status [1]; standard introductions explain how the SM unifies quark, lepton, electroweak, and strong-interaction phenomena [2].

The success of the SM should not be confused with a completed proof that every field in the model is indivisible. In experimental science, the defensible statement is that a field behaves pointlike within a tested kinematic regime and within a declared class of alternatives. A constituent model, a finite form factor, a four-fermion contact term, or an excited-fermion state can each alter a scattering distribution. Conversely, an inaccurate parton distribution function (PDF), an incomplete detector response, a background mismatch, or an invalid effective-field-theory (EFT) expansion can create a similar

apparent departure. The question of the smallest building blocks is therefore a problem in discriminating hypotheses under finite resolution, not a request for a final ontological certificate.

This report constructs a research proposal rather than an experimental result. The Survey Agent freezes evidence and boundary conditions. The Idea Agent selects a cross-channel model-comparison direction. The ExperimentDesign Agent specifies a computational, `DESIGN_ONLY` protocol. The Author Agent turns those inputs into an IEEE-style research plan. The proposed Resolution-Ladder Model Comparison (RLMC) determines when a short-distance deviation has enough kinematic, theoretical, and systematic support to be called evidence against a pointlike-SM explanation, and when the appropriate conclusion is only a range-specific constraint or an unresolved ambiguity.

II. SCOPE AND DEFINITIONS

A. What elementary means in this proposal

Table I fixes the vocabulary. A field classified as elementary in the SM is not assigned a resolved spatial substructure in the model. A composite hadron is a bound state with quark and gluon degrees of freedom. A pointlike experimental result means that the selected observables do not require a finite-size, contact, or excited-state deformation within the stated sensitivity, nuisance assumptions, and energy range. It does not prove that no deeper dynamics exists at arbitrarily high momentum transfer.

The Pauli exclusion principle applies to identical fermions in the same quantum state; bosons may occupy a common state. This difference in quantum statistics does not decide whether a particle is composite. Similarly, gauge bosons mediate gauge interactions, but the scalar Higgs boson is not best described as a force carrier. These distinctions matter because the proposal links a precise particle-physics claim to an equally precise experimental test rather than reproducing a simplified chart of particles.

B. Known scope and open scope

The SM has passed stringent tests in many sectors. Higgs production and decay analyses, for example, have tested selected coupling patterns against SM expectations in specified collision samples [3]. Yet the framework does not include a quantum theory of gravity, and the minimal massless-neutrino formulation is inadequate for neutrino-mass phenomena. It also does not uniquely identify the origin of all flavor patterns or the microscopic nature of dark matter. Those facts motivate

TABLE I
PARTICLE CLASSIFICATION AND OPERATIONAL STATEMENTS USED IN THE PROPOSAL.

Object	Standard-Model role	Operational statement	Prohibited overstatement
Quarks and leptons	Matter fermion fields in three generations	Their scattering can be tested for pointlike behavior, contact terms, form factors, or excited-state signatures over a declared range.	Proven indivisible at all scales.
Gauge bosons	Quanta associated with strong, weak, and electromagnetic gauge interactions	Their couplings and propagator structures are tested in selected processes.	Every boson is a force carrier.
Higgs boson	Scalar excitation associated with electroweak symmetry breaking	Production and decay channels can test selected coupling patterns against SM expectations.	A gauge-force carrier.
Protons and neutrons	Composite hadrons with quark and gluon dynamics	Hadron structure and PDFs must be modeled in a collider prediction.	Elementary particles.
EFT coefficient or form factor	Parameterization of an allowed deviation at finite resolution	It is a testable deformation with a cutoff and validity domain.	Direct proof of a unique constituent.

searches for departures, but they do not predict that a particular quark or lepton must be composite.

RLMC is deliberately narrower than a generic search for all beyond-SM physics. It focuses on finite-resolution signatures most directly connected to fermion pointlikeness: high-mass and high-transverse-momentum scattering tails, angular distributions, possible form-factor suppressions or enhancements, and benchmark excited-fermion spectra. It treats weakly coupled hidden-sector searches and low-energy precision probes as complementary context rather than direct measurements of quark radius [9], [10].

III. EVIDENCE SURVEY: FROM PARTICLE INVENTORY TO RESOLUTION TESTS

A. Hadronic structure and finite resolution

The mass and charge distribution of a proton are not evidence that its quarks can be isolated as free particles. In QCD, quark and gluon degrees of freedom are confined into color-neutral hadrons. Collider predictions therefore require a factorized but imperfect connection between partonic scattering and hadronic observables. PDFs summarize the momentum distributions of partons in a hadron for a stated scheme and scale. This makes them indispensable to high-energy tests of quark and lepton pointlikeness: a deformation in a high-mass spectrum can resemble a PDF shift unless the uncertainty model is explicit.

Lepton-hadron scattering is complementary because it can probe partonic structure with different systematics and kinematic lever arms. Proposed electron-ion measurements aim to improve understanding of hadronic and mesonic structure through broad kinematic coverage, but their performance estimates are not observed data [7]. RLMC uses such channels only as future or optional interfaces and labels them as projected until a released measurement and response model are available.

B. Contact interactions, form factors, and excited states

When unknown heavy physics cannot be produced directly in a given energy range, an EFT can summarize its leading

effects using higher-dimensional operators:

$$\mathcal{L}_{\text{eff}} = \mathcal{L}_{\text{SM}} + \sum_i \frac{C_i}{\Lambda^2} \mathcal{O}_i + \sum_j \frac{D_j}{\Lambda^4} \mathcal{O}_j + \cdots \quad (2)$$

The coefficient C_i , scale Λ , and operators $\mathcal{O}_i$ encode a low-energy deformation; they do not specify a unique ultraviolet constituent model. Dimension-eight coefficients D_j further show why a dimension-six-only conclusion can depend on an assumed truncation. A systematic SMEFT parameterization requires explicit choices of basis, flavor structure, and electroweak input scheme [5].

High-energy Drell–Yan tails can constrain semileptonic interactions, but the inferred reach depends on flavor assignments, higher-order operators, and the validity of the EFT at the momentum transfer actually used [4]. A finite-size interpretation can alternatively be expressed through a form-factor expansion:

$$F_f(Q^2) = 1 - \frac{r_f^2 Q^2}{6} + \mathcal{O}(Q^4), \quad (3)$$

where Q is a declared momentum-transfer variable and r_f^2 is a model-dependent radius parameter. This expansion must not be extrapolated past its stated regime. Excited-fermion searches supply another alternative with a different predicted shape. A recent excited-tau search reported no excess in its selected final state and set benchmark-dependent limits [6]; it did not establish that all charged leptons lack constituents at every scale.

C. Complementary resolution programs

Future colliders may offer both higher energy reach and precision control. The Future Circular Collider program describes complementarity among electron, proton, and ion collision modes, but projected sensitivities depend on machine and detector assumptions [8]. A resolution ladder should not append projected curves to existing data as if all were comparable measurements. It must retain an input-status field: observed public measurement, validated public-likelihood re-interpretation, synthetic closure test, or future projection. This status is part of the evidence rather than a formatting detail.

IV. RESEARCH GAPS AND IDEA SELECTION

The Survey Agent identifies six linked gaps. Resolution is not consistently defined across channels. Contact interactions are too readily described as literal compositeness without an EFT boundary. PDF and response-model systematics can confound a short-distance signal. Form-factor, contact, and excited-state alternatives are often analyzed in isolation. Collider, lepton-hadron, and precision programs need an explicit transfer contract rather than loose complementarity claims. Finally, many interpretation claims lack a versioned record of binning, response, theory, and decision conditions.

The Idea Agent compared a single-channel Drell–Yan fit, an excited-lepton scan, a lepton-hadron transfer study, and an integrated framework. The selected idea is RLMC because it does not merely add sources. It imposes a common inference contract: each observable receives a momentum-transfer definition, a status label, a response model, a nuisance ledger, and an EFT or non-EFT validity condition before model comparison begins.

The central hypothesis is that a shared resolution ledger and cross-channel likelihood will separate coherent substructure-like patterns from EFT-truncation and channel-specific systematic artifacts more reliably than independent single-channel limits. The primary falsifiers are equally clear. The program fails to support an alternative if model families are non-identifiable, if cutoff or operator-order changes reverse the preference, if nuisance perturbations reproduce the apparent departure, or if a proposed common coefficient pattern is incompatible across channels without an explicit ultraviolet model.

V. EXPERIMENT DESIGN

A. Research object, channels, and variables

The ExperimentDesign Agent selects a computational-digital template. The scientific object is not a physical collider operation but a finite-resolution inference over scattering observables. The baseline hypothesis is a pointlike-SM prediction. Alternative families include specified dimension-six and dimension-eight contact operators, finite form-factor deformations, and excited-fermion benchmarks. Neutral- and charged-current Drell–Yan-like observables form the primary planned interfaces because their high-scale tails and angular structure are relevant to semileptonic interactions [4]. Lepton-hadron and future-facility channels are optional and are explicitly labeled as synthetic or projected until independently approved public inputs exist.

The independent variables are momentum transfer or invariant mass Q , angular variable c , EFT coefficient ratios C_i/Λ^2 , declared dimension-eight terms D_j/Λ^4 , form-factor parameters r_f^2 , excited-state benchmark parameters, and nuisance vector η . The dependent objects are binned yields, normalized angular distributions, likelihood ratios, held-out predictive scores, and stability diagnostics. Controlled objects include binning version, theory order, PDF ensemble, fiducial definition, detector response, luminosity treatment, and data-release provenance.

B. Generative model and likelihood

For a channel c and bin b , RLMC records an expected yield

$$\mu_{cb}(\theta, \eta) = \mathcal{L} \int_b \left[R_{cb} \frac{d\sigma_c(\theta, \eta)}{d\Phi} \right] d\Phi B_{cb}(\eta). \quad (4)$$

Here $\mathcal{L}$ is integrated luminosity when a realized measurement is used, R_{cb} is the response operator, $d\sigma_c/d\Phi$ is the differential prediction in phase space Φ , B_{cb} is the background contribution, θ collects physics parameters, and η collects nuisance parameters. For synthetic closure tests, all terms are simulated inputs and are never described as observed data.

The joint likelihood is

$$\mathcal{L}(\theta, \eta) = \prod_{c,b} \text{Poisson}(n_{cb} \mid \mu_{cb}(\theta, \eta)) \prod_k \pi_k(\eta_k), \quad (5)$$

where n_{cb} is a binned record only in an independently approved interface and π_k represents declared nuisance constraints. The construction must preserve which nuisance parameters are correlated across channels and which are not. It is invalid to merge a future projection with a released measurement merely because both use the word sensitivity.

C. EFT validity and cross-channel contract

For a contact interaction, the inference includes a dimensionless diagnostic

$$\epsilon_{\text{EFT}} = \frac{Q_{\text{max}}}{\Lambda_{\text{cut}}}, \quad (6)$$

with a predeclared acceptance policy rather than a universal numerical threshold. The analysis repeats the comparison over a cutoff sweep and an alternative operator-order policy. A point estimate at one cutoff is not sufficient evidence for compositeness when the conclusion is unstable under these choices.

Cross-channel consistency is tested only when the assumed model supplies a relation among coefficients or shape parameters. Channels may have distinct PDFs, response effects, and backgrounds, so RLMC shares a nuisance object only when physics and calibration provenance justify it. This avoids a false appearance of precision from forcing unrelated channels into one covariance matrix.

VI. ANALYSIS AND DECISION RULES

A. Identifiability before preference

The central question is not whether an alternative can fit a flexible curve. It is whether pointlike, contact, form-factor, and excited-state families have distinguishable predictions over the declared bins after nuisance variation. RLMC begins with synthetic closure tests generated from each family. It measures profile degeneracies, posterior correlations, and the ability to recover an injected class under held-out bins. If a finite form factor and a response-model deformation have indistinguishable effects, a radius interpretation is non-identifiable and must not be reported as a compositeness indication.

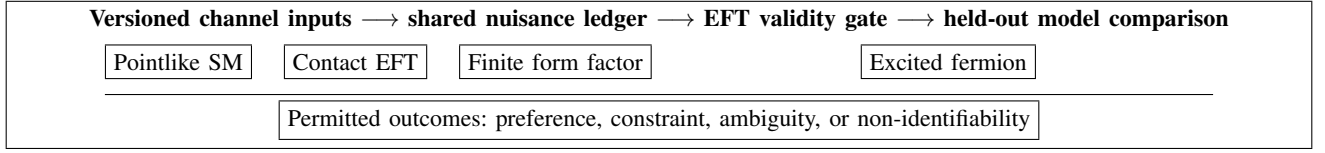


Fig. 1. RLMC inference architecture. Competing hypotheses are compared only after each channel’s kinematics, response, and nuisance treatment are versioned and after the EFT validity policy is applied. The editable Draw.io source is retained in the report project.

TABLE II
DESIGN_ONLY MODULES OF RESOLUTION-LADDER MODEL COMPARISON.

Module	Inputs and controls	Planned output	Failure or review condition
Channel contract	Momentum-transfer definition, binning, fiducial selection, status label, and response interface	Versioned channel manifest	Missing selection, incompatible Q definition, or projection presented as data
Pointlike baseline	SM prediction order, PDFs, scale variations, backgrounds, and nuisance priors	Baseline expected yields and uncertainty envelope	Prediction provenance or nuisance correlations unavailable
Contact-EFT alternatives	Operator basis, coefficient set, flavor assumptions, cutoff, dimension-eight policy	EFT likelihood and validity diagnostic	Cutoff or truncation sensitivity changes the conclusion
Finite-size alternative	Declared form-factor expansion and range of use	Radius-like shape template	Expansion used outside justified kinematics or degenerate with response effects
Excited-state alternative	Mass, coupling, and compositeness-scale benchmark	Resonant or nonresonant template family	Benchmark not distinguishable from contact or background shape
Closure and holdout	Synthetic data, adversarial nuisance injection, training/hold-out bins	Identifiability and predictive comparison	A nuisance model reproduces an apparent preference

The primary comparison statistic is

$$\Delta T = -2 \log \left[\frac{\mathcal{L}(\hat{\theta}_{\text{pointlike}}, \hat{\eta}_{\text{pointlike}})}{\mathcal{L}(\hat{\theta}_{\text{alternative}}, \hat{\eta}_{\text{alternative}})} \right]. \quad (7)$$

Equation (7) does not establish coverage or model preference by itself. Its calibration must be checked with the appropriate pseudoexperiment, asymptotic, or Bayesian procedure selected by a qualified analysis team. The statistic is paired with held-out predictive performance, cutoff variation, and nuisance stress rather than interpreted in isolation.

B. Stress tests and decision branches

The first stress test varies the PDF ensemble and its correlations. The second perturbs detector response, background normalization, and luminosity terms within their declared uncertainty model. The third changes binning and withholds high- Q bins. The fourth changes EFT order, operator subset, flavor assumption, and cutoff. The fifth injects controlled contact, form-factor, and excited-state signals into synthetic closure records. A conclusion is stable only if the relevant model remains identifiable and its improvement persists under predetermined stress tests.

C. Reproducible reporting

Every run produces a channel manifest, data-status label, theory calculator version, selection and binning record, nuisance ledger, EFT policy, synthetic closure seed, and decision log. The manifest is more than administrative metadata. It permits another analyst to identify whether a comparison used a real public measurement, a public likelihood recast,

TABLE III
PREDECLARED RLMC DECISION BRANCHES.

Condition	Permitted conclusion
Identifiable, cutoff stable, nuisance robust, and cross-channel coherent alternative	The selected alternative is preferred within its declared model class and tested resolution range. It is not proof of a unique constituent.
Identifiable alternative with no predictive gain	The pointlike baseline remains adequate for tested observables and assumptions. Report a range-dependent constraint.
No hypothesis identifiability	No source-specific or constituent interpretation is warranted; revise bins, observables, or calibration.
EFT-order or cutoff instability	Do not quote a compositeness interpretation from that region; restrict or change the model.
Nuisance injection mimics deviation	Treat the result as a systematic-model ambiguity, not new physics.

a synthetic record, or a future projection. This is essential when integrating SMEFTsim-like coefficient calculators [5] with external experimental inputs.

VII. EXPECTED OUTCOME BRANCHES AND SIGNIFICANCE

If RLMC finds that a pointlike-SM baseline is adequate under a validated public or future experimental interface, the correct result is a stronger and more transparent range-dependent constraint. It specifies the scales, channels, alternatives, and nuisance assumptions for which no additional substructure is required. Such a result does not close the possibility of deeper degrees of freedom beyond accessible resolution.

If an alternative is identifiable and remains preferred after the full stress program, the correct result is a model-conditioned anomaly or preference. It establishes neither a preon nor a final constituent. A contact coefficient can arise from many ultraviolet mechanisms; a form factor can have modeling ambiguities; an excited-state signal requires its own background and reconstruction tests. The next stage would be an explicit ultraviolet-model comparison and independent channel confirmation, not a statement that the smallest building block has been found.

If the analysis is non-identifiable, the negative result is useful. It identifies which nuisance or theory assumptions prevent a meaningful conclusion and motivates a better observable, a different kinematic regime, an improved response model, or a complementary facility. This is where high-energy, lepton-hadron, precision, and low-energy programs become strategically complementary [8]–[10].

VIII. LIMITS AND HUMAN REVIEW REQUIREMENTS

The proposal has substantive limits. PDFs are not optional details in proton-collision interpretations. A form-factor expansion has a finite domain of validity. EFT truncation is not eliminated by writing one operator in Eq. (2). An excited-state benchmark is not a complete compositeness theory. Cross-channel correlation is physically meaningful only when inputs and model justify it. A symmetry-respecting extension of the SM can parameterize possible departures, but parameterization is not evidence that a particular departure exists [11].

The workflow is also operationally limited. It does not operate accelerators, steer beams, configure triggers, control detectors, access restricted collider records, or publish a physics limit. Before any public-data application, a collider phenomenologist and experimental analyst must confirm data-release terms, fiducial definitions, detector response, likelihood validity, theory order, and statistical coverage. Before an external submission, a human must verify publisher or DOI landing pages because the requested application browser could not run in the Windows ACL sandbox.

IX. CONCLUSION

The Standard Model currently provides the best-tested inventory of elementary matter fields: quarks and leptons, together with gauge and Higgs bosonic fields. Protons and neutrons are composite hadrons, not the final building blocks. The experimentally honest conclusion is that known quarks and leptons behave pointlike over tested scales and assumptions, not that every possible deeper layer has been ruled out forever.

RLMC turns this distinction into a concrete research program. It compares pointlike-SM, contact-EFT, finite-form-factor, and excited-fermion hypotheses across versioned channels, with an explicit nuisance ledger and EFT-validity gate. It treats a null result, a model preference, a systematic ambiguity, and non-identifiability as different scientific outcomes with different permitted conclusions. It asks not only what model

is currently adequate, but also what evidence would genuinely require us to refine it.

X. RESOLUTION ROAD MAP AND EVIDENCE UPGRADING

A. A staged program rather than a single limit

The question of pointlike behavior is best treated as a staged program. The first stage is synthetic closure. It asks whether the selected channel, binning, response model, and nuisance ledger can recover a known injected pointlike, contact, form-factor, or excited-state pattern. This is a test of the inference apparatus, not a stand-in for a collision measurement. It is especially valuable because a fitted contact coefficient can otherwise be driven by a poorly modeled tail or a hidden response correlation.

The second stage is channel qualification. A channel enters the common comparison only after its kinematic definition, fiducial selection, theory calculator, response model, and data status are registered. A public binned measurement without a response interface can be included as a qualitative evidence source but not as a quantitative likelihood. A public likelihood can be used only under its documented validity conditions. A future collider or electron-ion facility remains a projection until it produces released data. This staged admission protects the program from assigning equal weight to incomparable inputs.

The third stage is cross-channel synthesis. The combined analysis should not ask whether every channel has the same apparent deviation. Instead, it asks whether a particular physics model predicts a consistent coefficient, form-factor scale, or excited-state benchmark after the channels retain their own PDFs, backgrounds, and response effects. A disagreement may rule out the selected common model, expose an interface error, or point to a more complicated alternative. It does not become evidence for compositeness by itself.

B. How each outcome changes the scientific answer

The pointlike-SM branch is not an empty outcome. If it survives the qualified likelihood comparison, it supports a sharper statement: within the registered momentum-transfer interval, fiducial definition, uncertainty model, and alternative class, no additional fermion substructure is required. This statement can be compared across runs because the resolution ledger records the terms under which it is true. It remains stronger than a casual assertion that nothing was found, yet it does not outgrow its evidence.

The contact-EFT branch has a different interpretation. A stable contact coefficient can identify a departure from the pointlike baseline in an effective description, but it cannot by itself identify a constituent scale or a unique ultraviolet completion. The analysis must disclose whether the preferred coefficient is robust to changes in cutoff, dimension-eight terms, and flavor assumptions. High-energy tail studies make this issue explicit: the utility of the tail is matched by sensitivity to the domain in which the EFT description is reliable [4].

TABLE IV
RESOLUTION-LADDER EVIDENCE-UPGRADING RULES.

Stage	Admissible input	Primary question	Go or no-go rule
Closure	Synthetic records generated from stated hypotheses and nuisance models	Can the inference recover the injected class and separate it from adversarial nuisance shifts?	No-go for quantitative interpretation if classes are non-identifiable.
Qualification	Released binning, selections, response information, and theory provenance	Is the channel definition sufficient to construct a reproducible likelihood?	No-go for combined fit if status or response provenance is missing.
Single-channel fit	Pointlike and alternative predictions under a declared cutoff policy	Does any alternative improve held-out prediction without a nuisance explanation?	Report only a channel-conditioned preference or constraint.
Cross-channel synthesis	Qualified channels connected by an explicit model relation	Is a shared alternative compatible with all relevant channel likelihoods?	No-go for a common interpretation if coefficient or shape relations conflict.
Independent confirmation	A separate observable or facility with materially different systematics	Does the preferred pattern survive a new resolution and uncertainty environment?	Upgrade only after independent confirmation and repeated validity tests.

The form-factor branch asks a still different question. A radius-like parameter can summarize a smooth momentum-transfer dependence in a valid low- Q expansion. Its apparent value must be tested against response functions and PDF variations that can also tilt a spectrum. An excited-fermion branch adds resonance, acceptance, and reconstruction questions. Existing excited-lepton searches illustrate why a benchmark limit must retain its mass and coupling assumptions [6]. RLMC keeps those hypotheses separate because their failure modes are separate.

C. Strategic value of complementary facilities

Energy, luminosity, flavor reach, and experimental systematics are not interchangeable resources. A future hadron collider can extend high- Q reach; a lepton collider can emphasize cleaner initial conditions and precision; an electron-ion program can constrain partonic structure with a different kinematic organization. Their complementarity is valuable only when it is expressed through a transfer condition and status label, not through a decorative collection of sensitivity plots. Planning studies for future colliders and electron-ion measurements provide motivation for these distinct roles [7], [8].

The resulting scientific answer is dynamic but rigorous. The smallest currently tested fields are those whose pointlike-SM descriptions remain adequate after the best available resolution tests and stated alternatives. A future result may lower a radius bound, exclude an excited-state region, restrict an EFT coefficient, or reveal a coherent deviation. Each possibility refines the answer without requiring the proposal to claim that it has already occurred.

APPENDIX A PROVENANCE AND EXECUTION CONTRACT

The proposal inherits Survey identity survey-elementary-matter-001, project identity phys_7_elementary_matter_20260901, and the version-1 context of Standard-Model pointlike-fermion compositeness resolution. The Idea Agent selects idea-rlmc-001; the ExperimentDesign Agent binds it to a computational-digital design.

The execution policy is DESIGN_ONLY. Observed results are an empty set, automatic execution is disabled, and human review is required. Planned synthetic closures, future-facility estimates, and public-data interfaces are marked as distinct statuses. This prevents the research plan from inventing collision observations, an exclusion scale, or a discovery claim.

APPENDIX B CLAIM-EVIDENCE MATRIX

For every later implementation, the record must declare the channel and data status; Q definition and binning; fiducial selection; response and background model; theory calculator and PDF prescription; nuisance correlations; EFT basis, cut-off, and truncation policy; alternative-hypothesis parameters; closure-test configuration; held-out split; and decision branch from Table III. This is the minimum audit trail from a finite-resolution observation to a bounded conclusion.

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TABLE V
CLAIM–EVIDENCE MATRIX. ALL CITATIONS ARE FROZEN BY THE SURVEY AGENT.

Bounded report claim	Evidence keys	Inference constraint
The SM classifies quark, lepton, gauge, and Higgs fields while leaving open scope.	pdg2008, cottingham2007, cms_higgs2015	Do not transform a successful inventory into proof against all high-scale alternatives.
High-energy Drell–Yan tails can constrain semileptonic EFT interactions.	allwicher2023, brivio2017	Declare flavor assumptions, operator basis, EFT order, and cutoff policy.
Excited-fermion null searches give benchmark-dependent constraints.	cms_tau2025	Do not make a global indivisibility claim from one channel or benchmark.
Lepton-hadron and future-collider studies are complementary.	arrington2021, abada2019	Preserve observed, synthetic, and projected status labels.
Low-energy and weakly coupled searches motivate a broader program.	alekhin2016, jaeckel2010, colladay1998	Do not present a parameterization or search motivation as detection evidence.

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